

Mohak Kapoor

ML Engineer focused on end-to-end machine learning systems on edge and cloud

🌐 Portfolio ✉ contact.mohakapoor@gmail.com ☎ +91 98702-32056 in Mohak Kapoor 🌐 mohakapoor

Technical Skills

Languages	Python (Advanced), C++
ML & Modeling	Classical ML (Regression, SVMs, Tree-based models), Time-Series Forecasting, CNNs, RNNs
Systems & Deployment	REST APIs, Docker, CI/CD (GitHub Actions), PostgreSQL, Edge Deployment on Raspberry Pi5
Libraries / Frameworks	TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, Polars

Projects

Network Anomaly Detection [↗](#)

Sep-Dec 2025

- Built and deployed an end-to-end intrusion detection system from multi-day CICIDS-2017 traffic to live inference, designed for edge deployment on resource-constrained hardware (Raspberry Pi 5).
- Achieved 96.7% test accuracy ($F1 = 0.968$) for binary attack detection using SVM and 99% multiclass accuracy ($F1 = 0.98$) using LightGBM, with time-aware train/test splits to reflect real deployment conditions.
- Operationalized the full ML pipeline as a REST API on Raspberry Pi 5, demonstrating real-time inference without GPU acceleration via a web interface under constrained compute and memory.

CaptchaOCR — CAPTCHA Recognition System [↗](#)

- Built an end-to-end CAPTCHA recognition pipeline using a CNN + BiLSTM (CRNN) architecture with CTC loss, handling variable-length and distorted character sequences.
- Achieved 96%+ character-level accuracy through synthetic data generation and targeted preprocessing, improving robustness to noise and deformation.
- Deployed the trained model on Hugging Face for inference, enabling public access and validation beyond offline experimentation.

Solar Power Generation Predictor [↗](#)

- Developed a TensorFlow model to predict solar power output using OpenWeatherMap API, geographical, and time-based features, hosted on a Django web platform.
- Tools: Python, sklearn, TensorFlow, Pandas, NumPy, Django, OpenWeatherMap API

Experience

HumanizeIQ — AI Intern Integrations

June-Oct 2025

- Architected and deployed production MCP backend services using Cloudflare Workers to support asynchronous image generation and multimodal document workflows, with persistent storage via R2.
- Operationalized end-to-end recruiter call analysis pipelines by integrating LLMs (Gemini, GPT-4) for transcription analysis, automated report generation, and downstream email delivery, eliminating manual post-call processing.
- Built and exposed REST APIs for background job orchestration and result retrieval, enabling reliable execution of long-running AI tasks and seamless integration with chat-based LLM systems.

JPMorgan Chase & Co. Quantitative Research Virtual Experience

Jan 2025

- Analyzed a consumer loan portfolio to estimate probability of default, applying quantitative reasoning and dynamic programming to bucket FICO scores, improving robustness of downstream risk modeling.

Education

Jaypee Institute of Information Technology, Noida-62

Sept 2022 – July 2026

B.Tech in Electronics and Communication Engineering

- **Relevant Coursework:** Machine Learning (A+), Deep Learning (A+), Advanced Statistics (A), Computer Systems, Multivariate Calculus, Differential Calculus, Integral Calculus, Control Systems.